

Can Frontier LLMs One-Shot Translate Lakota? (Mostly No.)

Evaluating Frontier LLM Translation Capability for Lakota

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Sixth Workshop on NLP for Indigenous Languages of the Americas (AmericasNLP 2026)

No frontier LLM reliably translates Lakota.

Even measuring how badly they fail is hard.

Lakḥótiyapi

the language under test

Why test this?

Lakḥótiyapi (Lakota) is a Siouan language UNESCO classifies as **critically endangered** — fewer than 2,000 first-language speakers, most over 65.

Today’s LLMs list translation support across dozens of languages, but providers don’t say what they can or can’t do per language. A reasonable user can’t tell Lakota apart from any of the others on the list — *if it does all these, why not this one?* Direct evaluation is the only way to find out.

Is it just memorized?

The evaluation set comes from a published dictionary, so contamination is the obvious worry. The Lakota source appears verbatim online for ~10% of pairs, but the Lakota and its English reference together for only 0.5%. Scores spread widely (6–61% equivalence), not clustered at ceiling — the signature of partial capability, not recall.

Setup

- **7 models.** Proprietary: Gemini 3.1 Pro, Claude Opus 4.6, Claude Sonnet 4.6, GPT-5.2. Open-weight: DeepSeek-V4-Pro, Qwen3.6-Plus, GLM-5.1.
- **200 sentence pairs** from the New Lakota Dictionary, conversational register, **both directions**.
- **Two conditions:** baseline vs. extended reasoning, where the API permits.
- **Metrics:** chrF + + and BLEU; diacritic-normalized chrF + +; an LLM *semantic* judge on the English side.

Why two judges?

chrF + + rewards surface overlap; it cannot tell fluent-but-wrong from correct. Meaning needs a separate check. For Lakota → English the model output *is* English, so judging meaning takes no Lakota at all — that’s what made an LLM judge feasible.

To keep one model from flattering its own family, every pair is judged by **two models from different families** — Gemini 3 Flash and the independent Llama-3.3-70B — which agree substantially: Cohen’s $\kappa = 0.75$ (0.89 quadratic-weighted).

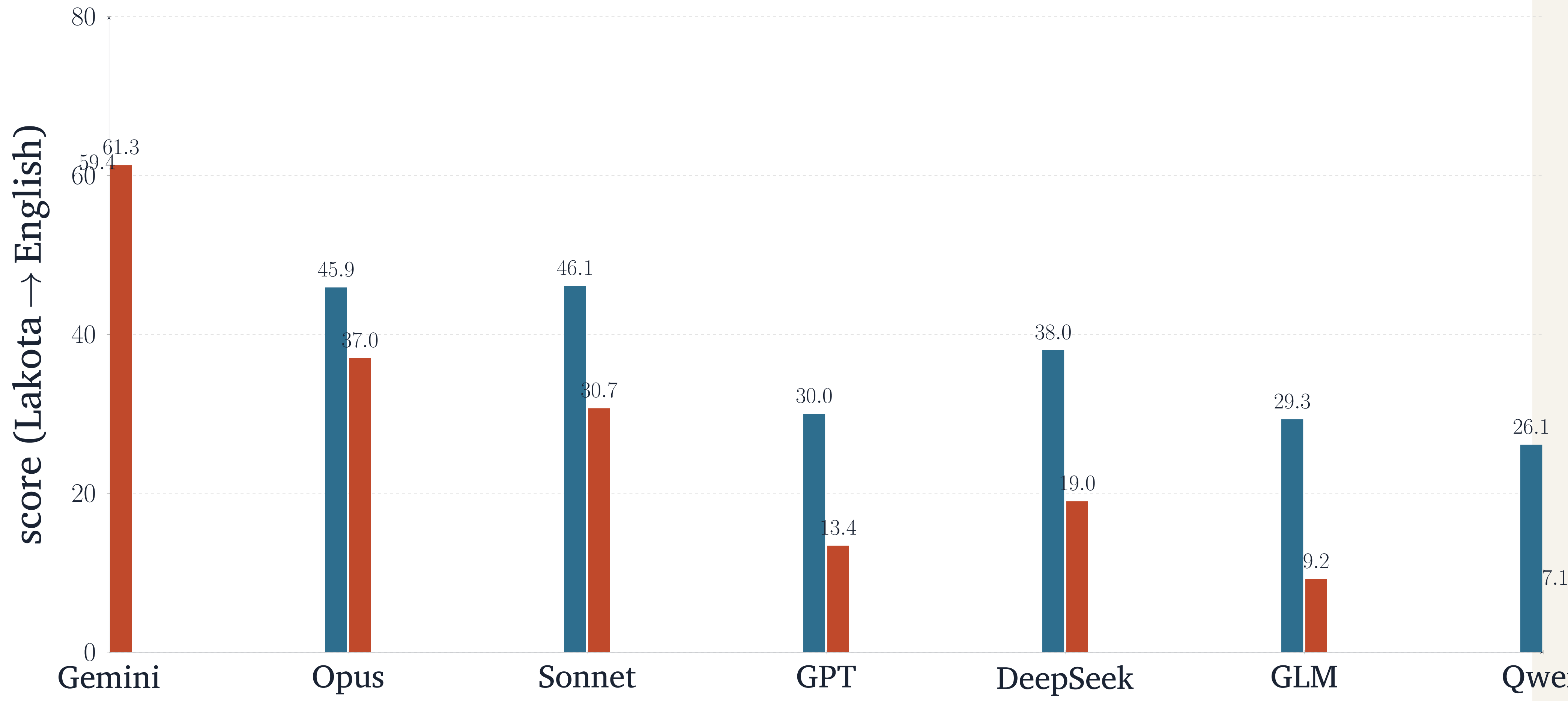
Standard Lakota Orthography

ḥ q ṭ ȩ á ŋ ’

SLO encodes phonemic distinctions with diacritics; stripping them adds +3.3 to +6.5 chrF + + (E → L) — models get base characters right, marks wrong.

Surface metrics overstate capability

chrF + + % semantically equivalent



The picture shows the divergence; the tables below prove it. chrF + + stays in the 30s–40s even where genuine semantic equivalence collapses below 15%. The two metrics agree for the strongest model and diverge sharply for the rest — so chrF + + alone is misleading for low-resource translation.

Translation quality (chrF++)

Model	L → E	E → L
Proprietary		
Gemini 3.1 Pro	59.4	42.6
Claude Opus 4.6	45.9	34.3
Claude Sonnet 4.6	46.1	27.0
GPT-5.2	30.0	23.6
Open-weight		
DeepSeek-V4-Pro	38.0	29.2
GLM-5.1	29.3	14.6
Qwen3.6-Plus	26.1	18.4

Best condition per model. Every model scores higher *out of* Lakota than *into* it.

How models fail

Reference	Model	Output	chrF + +
We live in a small trailer house with no electricity.	DeepSeek	“small <i>sacred</i> tipi”	18.6
Make coffee (<i>Wakḥálya</i> yo).	DeepSeek	“Sanctify it!”	2.9
Did my daughter call?	GLM	“Did my younger brother come?”	27.6
Do you have a car?	GPT-5.2	“Do you have a horse?”	66.4

Idioms and coined terms are translated *element by element*: *Wakḥálya yo* (“make coffee,” literally “make it boil”) becomes “sanctify it.” A one-word near-miss — “car” rendered “horse” by a *frontier* model — still scores 66.4 chrF + +. Surface overlap and meaning preservation diverge.

Semantic equivalence (L → E)

Model	Equiv.	Partial	Not
Gemini 3.1 Pro	61.3	23.1	15.6
Claude Opus 4.6	37.0	21.0	42.0
Claude Sonnet 4.6	30.7	23.8	45.5
DeepSeek-V4-Pro	19.0	14.0	67.0
GPT-5.2	13.4	9.8	76.8
GLM-5.1	9.2	6.5	84.2
Qwen3.6-Plus	7.1	7.6	85.4

% of judged pairs, reasoning condition.

Takeaways

- No frontier LLM reliably translates Lakota, either direction (2026).
- chrF + + overstates low-resource capability — pair it with a meaning check.
- Open-weight doesn’t close the gap.

Extended reasoning changes refusal behavior more than quality (open-weight E → L: **0** → **36** refusals when enabled).

Origin

This evaluation began as a response to a provocation from Benjamin Bratton (UCSD): “*just vibe-code a Lakota translation app over the weekend.*” We built the evaluation that would test whether such an app could exist.



Data & code

github.com/robotson/
lakota-translation-benchmark